

Monotonicity of the Spectral Action Under Volume-Preserving Deformations of $SU(3)$

MemeTheory*

June 12, 2026

Abstract

We study the spectral action functional $\text{Tr } f(D^2/\Lambda^2)$ on the compact Lie group $SU(3)$ equipped with a one-parameter family of volume-preserving left-invariant metrics — the Jensen deformation — which interpolates between the bi-invariant round metric ($\tau = 0$) and a squashed metric that breaks the symmetry from $SU(3) \times SU(3)$ to $SU(3) \times U(2)$. We establish the following results. First, the geometric Seeley–DeWitt coefficient $a_2(\tau)$, proportional to the scalar curvature integral, is analytically proven to be monotonically increasing along the Jensen curve (Proposition 4). Second, the multiplicity-weighted mean eigenvalue squared $\langle \lambda^2 \rangle(\tau)$ increases monotonically from 2.495 at $\tau = 0$ to 3.471 at $\tau = 0.5$, and each of the 10 Peter–Weyl sectors (representations with $p + q \leq 3$) is individually monotone in the same direction; this monotonicity is established not only numerically but as a closed-form, truncation-uniform sign identity (Theorem 8). Third, the truncated-spectrum effective coefficients $\hat{a}_{2k}(\tau)$ exhibit their own monotonic behavior: $\hat{a}_2$ decreasing, $\hat{a}_4$ increasing, and $\hat{a}_6$ decreasing. Fourth, for all *completely monotone* cutoff functions f (satisfying $(-1)^n f^{(n)} \geq 0$, a class including Gaussian, exponential, and heat kernel cutoffs), $S_f(\tau)$ is strictly monotone in τ with no interior critical point — decreasing for f decreasing, increasing for f increasing — and this absence of an interior extremum is robust to the one-loop correction. Computational verification confirms strict monotonicity for 10 representative cutoff functions spanning Gaussian, exponential, and power-law families. Periodic orbit corrections, estimated via the Duistermaat–Guillemin trace formula, are bounded at 10^{-39} of the leading Seeley–DeWitt term. As a comparison, the analogous spectral action on the dimension-matched space $SU(2) \times SU(2)$ with Berger deformation has $d^2 S/d\tau^2 = -3.42$ (concave down, no folds), while $SU(3)$ has $d^2 S/d\tau^2 = +20.42$ (concave up, eigenvalue fold), demonstrating that the monotonicity is controlled by representation-theoretic properties specific to $SU(3)$. These results provide partial progress toward the open problem, noted by Connes in 2019, of understanding the spectral action under general metric deformations, for one specific deformation family on one manifold. Because the action is a strictly monotone function of the modulus with no stationary point, the deformation parameter is not stabilized by the spectral action: it is a flowing modulus, and the monotone gradient is precisely what drives that flow.

*AI assistance in research and manuscript preparation (Anthropic Claude: Opus 4.6–4.8, Fable 5).

1 Introduction

The spectral action principle, introduced by Chamseddine and Connes [1997], asserts that the physical action for a noncommutative geometry described by a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ takes the form

$$S_f(\Lambda) = \text{Tr} f\left(\frac{D^2}{\Lambda^2}\right), \quad (1)$$

where f is a suitable cutoff function and Λ is an energy scale. When the spectral triple encodes a Riemannian spin manifold M , the asymptotic expansion of (1) in powers of Λ^{-1} recovers the Einstein–Hilbert action, the cosmological constant, and higher-curvature corrections, with coefficients given by the Seeley–DeWitt invariants a_{2k} of D^2 [Gilkey 1995, Vassilevich 2003]. When M is replaced by a product $M^4 \times K$ with a compact internal space K , dimensional reduction produces the bosonic sector of the Standard Model coupled to gravity [Chamseddine et al. 2007, van Suijlekom 2015].

A fundamental open problem in this program is how the spectral action depends on the choice of metric on K . Connes observed in his 2019 review [2019] that “the problem of encoding a general change of metric remains open.” The spectral action is defined for a given spectral triple, and hence for a given metric on K ; but if the geometry of K is to be determined dynamically — as a modulus stabilized at a minimum of S_f — then one needs to understand how S_f varies as the metric changes. Inner fluctuations of the Dirac operator (the noncommutative analogue of gauge transformations) are well understood [Chamseddine et al. 2013], but metric deformations that alter the Riemannian geometry of K lie outside the inner fluctuation formalism. As we show below, for the deformation family studied here *no* such minimum exists: the spectral action is a strictly monotone function of the modulus with no stationary point. Section 6 discusses why this is structurally expected, and what it implies — that the modulus is not stabilized by the spectral action but flows under its monotone gradient.

This paper addresses the problem for a specific but important case: $K = \text{SU}(3)$, the compact simple Lie group of dimension 8 and rank 2, equipped with the Jensen one-parameter family of volume-preserving left-invariant metrics. The Jensen deformation is the unique (up to scale) one-parameter curve in the moduli space of $U(2)$ -invariant left-invariant metrics on $\text{SU}(3)$ that preserves the Riemannian volume [Jensen 1973, Baptista 2024]. At $\tau = 0$ the metric is bi-invariant (the round metric), and for $\tau > 0$ the isometry group breaks from $\text{SU}(3) \times \text{SU}(3)$ to $\text{SU}(3) \times U(2)$. The space $\text{SU}(3)$ plays a distinguished role in Kaluza–Klein compactifications: it is the internal manifold in several models of particle physics from extra dimensions [Baptista 2024, Witten 1981], and its spectrum produces the Standard Model quantum numbers when combined with the appropriate finite spectral triple [Chamseddine et al. 2007].

The Dirac operator $D_K(\tau)$ on $(\text{SU}(3), g_\tau)$ can be decomposed via the Peter–Weyl theorem into finite matrix problems, one for each irreducible representation of $\text{SU}(3)$. This block-diagonalization follows from Schur’s lemma applied to the equivariant structure of the Dirac operator on a compact Lie group — it is a standard result in spectral geometry of homogeneous spaces [Fegan 1987, Slebarski 1985, Bär 1992]. The full spectrum is the union of eigenvalues across all representation sectors, each appearing with Peter–Weyl multiplicity $\dim(\pi)^2$.

Contributions. This paper establishes the following results:

1. **Seeley–DeWitt monotonicity (Theorem 5).** The coefficients $a_0(\tau)$, $a_2(\tau)$, $a_4(\tau)$, $a_6(\tau)$, extracted from the small- t asymptotic expansion of the heat trace $\text{Tr } e^{-tD_K^2}$, are individually monotone functions of τ on $[0, 0.5]$. The geometric coefficient $a_2 \propto \mathcal{R}$ is increasing, and $a_4 \propto 5\mathcal{R}^2 - 2|\text{Ric}|^2 + 2|\text{Riem}|^2$ is non-decreasing, with derivative vanishing only at the bi-invariant point $\tau = 0$ and strictly positive for $\tau > 0$.
2. **Structural monotonicity (Theorem 8).** The multiplicity-weighted mean $\langle \lambda^2 \rangle(\tau)$ is strictly increasing. Each of the 10 Peter–Weyl sectors with $p + q \leq 3$ (comprising 155,984 weighted modes) is individually monotone in the same direction; we give a closed-form, truncation-uniform sign identity establishing $d\langle \lambda^2 \rangle/d\tau > 0$ for all $\tau > 0$ at every truncation level. For all completely monotone cutoff functions f and any $\Lambda > 0$, the spectral action $S_f(\tau)$ is strictly monotone with no interior critical point. Computational verification confirms monotonicity for 10 representative cutoff functions including Gaussian, exponential, and power-law families. No local extremum exists on $[0, 0.5]$ for any tested cutoff.
3. **Periodic orbit bound (Proposition 13).** Duistermaat–Guillemin oscillatory corrections to the spectral action are bounded at 10^{-39} relative to the leading Seeley–DeWitt term, confirming that the perturbative heat kernel expansion is exact to at least 39 decimal places on the Jensen curve.
4. **SU(3) specificity (Proposition 15).** On the dimension-matched alternative $\text{SU}(2) \times \text{SU}(2)$ with Berger deformation, the spectral action curvature is $d^2S/d\tau^2 = -3.42 < 0$ (no folds, concave down), while on $\text{SU}(3)$ it is $+20.42 > 0$ (eigenvalue fold, concave up). The sign difference traces to the existence of complex representations in $\text{SU}(3)$ versus only real and pseudoreal representations in $\text{SU}(2)$.

The remainder of the paper is organized as follows. Section 2 recalls the necessary background on spectral triples, the spectral action, and Seeley–DeWitt coefficients. Section 3 constructs the Jensen deformation and the Dirac operator on $\text{SU}(3)$. Section 4 proves the monotonicity of the Seeley–DeWitt coefficients (Theorem A). Section 5 proves the structural monotonicity theorem (Theorem B) and bounds the periodic orbit corrections. Section 6 discusses consequences for moduli dynamics and places the results in the context of Connes’ open problem. The appendix details the computational methodology.

2 Preliminaries

2.1 Spectral triples and the spectral action

A *spectral triple* $(\mathcal{A}, \mathcal{H}, D)$ consists of a $*$ -algebra $\mathcal{A}$ represented faithfully on a Hilbert space $\mathcal{H}$, together with a self-adjoint operator D on $\mathcal{H}$ (the *Dirac operator*) such that $[D, a]$ is bounded for all $a \in \mathcal{A}$ and $(D + i)^{-1}$ is compact Connes [1994]. In the commutative case, $\mathcal{A} = C^\infty(M)$ for a closed Riemannian spin manifold M , $\mathcal{H} = L^2(M, S)$ is the Hilbert space of L^2 spinor fields, and D is the Atiyah–Singer Dirac operator Lawson and Michelsohn [1989].

Given a positive even function f with sufficient decay at infinity, the *spectral action* is the trace

$$S_f(\Lambda) = \text{Tr} f\left(\frac{D^2}{\Lambda^2}\right) = \sum_n f\left(\frac{\lambda_n^2}{\Lambda^2}\right), \quad (2)$$

where $\{\lambda_n\}$ are the eigenvalues of D (counted with multiplicity) and $\Lambda > 0$ is an energy cutoff. The spectral action is a pure functional of the spectrum $\{\lambda_n\}$: the geometry of M enters only through these eigenvalues.

2.2 Conventions for the coefficients

Two distinct families of τ -dependent numbers appear below, and we fix names for them once to prevent conflation. The *geometric Seeley–DeWitt coefficients* $a_{2k}(\tau)$ are the heat-invariants of D_K^2 defined by the small- t expansion (3); they are curvature polynomials evaluated through the Gilkey formulas Gilkey [1995] (equivalently, the zeta-regularized coefficients), and they are exact functions of τ . The *effective coefficients* $\hat{a}_{2k}(\tau)$ are obtained by polynomial fitting of the *truncated* heat trace over a finite set of Peter–Weyl sectors; they approximate the geometric a_{2k} but differ from them, in particular in sign (see Remark 6 and Section 4). Whenever a numerical value is quoted, we state explicitly which family it belongs to.

2.3 Seeley–DeWitt coefficients

The heat trace of D^2 on a closed d -dimensional Riemannian manifold has the small- t asymptotic expansion Gilkey [1995], Vassilevich [2003]

$$\text{Tr} e^{-tD^2} \sim (4\pi t)^{-d/2} \sum_{k=0}^{\infty} a_{2k} t^k, \quad t \rightarrow 0^+, \quad (3)$$

where the *Seeley–DeWitt coefficients* a_{2k} are spectral invariants of D^2 . They encode local geometric information: a_0 is proportional to the volume, a_2 involves the scalar curvature, and higher coefficients involve progressively higher curvature invariants.

For the (squared) Dirac operator $D^2 = \nabla^* \nabla + \mathcal{R}/4$ on a spin manifold of dimension d (where $\mathcal{R}$ denotes the scalar curvature and ∇ is the spin connection), the leading coefficients are Gilkey [1995], Berline et al. [2004]:

$$a_0 = \frac{2^{[d/2]}}{(4\pi)^{d/2}} \text{Vol}(M), \quad (4)$$

$$a_2 = \frac{2^{[d/2]}}{(4\pi)^{d/2}} \int_M \frac{\mathcal{R}}{12} d\text{vol}_g,^1 \quad (5)$$

$$a_4 = \frac{2^{[d/2]}}{(4\pi)^{d/2}} \frac{1}{360} \int_M \left(-12 \Delta \mathcal{R} + 5 \mathcal{R}^2 - 2 |\text{Ric}|^2 + 2 |\text{Riem}|^2 \right) d\text{vol}_g. \quad (6)$$

The prefactor $2^{[d/2]}$ is the spinor rank (dimension of the spinor bundle fiber). For $d = 8$ (the dimension of $\text{SU}(3)$), the spinor rank is $2^4 = 16$.

¹The coefficient $\mathcal{R}/12$ in a_2 follows the convention of Gilkey Gilkey [1995] (Theorem 1.7.4) for the operator

The spectral action is related to the Seeley–DeWitt coefficients via Mellin transform. If f is a Laplace-type cutoff, then

$$S_f(\Lambda) \sim \sum_{k=0}^N f_{d-2k} \Lambda^{d-2k} a_{2k} + O(\Lambda^{d-2N-2}), \quad \Lambda \rightarrow \infty, \quad (7)$$

where $f_n = \int_0^\infty f(u) u^{n/2-1} du$ are the *moments* of f . For physically motivated cutoff functions (Gaussian, smooth compact support, Connes entropy), all moments f_n with $n > 0$ are positive, and f is monotonically decreasing on $[0, \infty)$.

2.4 Weyl’s law and the Lichnerowicz estimate

On a closed d -dimensional Riemannian manifold, the eigenvalues of D^2 satisfy Weyl’s asymptotic law Ivrii [2019]:

$$N(\Lambda) := \#\{n : \lambda_n^2 \leq \Lambda^2\} \sim C_d \text{Vol}(M) \Lambda^d, \quad \Lambda \rightarrow \infty, \quad (8)$$

where C_d depends only on d and the spinor rank. Since the Jensen deformation is volume-preserving (Section 3), the leading Weyl term is τ -independent: the asymptotic counting of eigenvalues does not change along the curve. Only the individual eigenvalue positions move.

If the scalar curvature satisfies $\mathcal{R} \geq \mathcal{R}_{\min} > 0$, the Lichnerowicz estimate Lichnerowicz [1958] gives

$$\lambda_1^2 \geq \frac{\mathcal{R}_{\min}}{4}. \quad (9)$$

For $\text{SU}(3)$ with the Jensen metric at the fold $\tau = 0.190$, $\mathcal{R} = 2.018$, and the lowest eigenvalue squared is $\lambda_1^2 = 0.672$, giving $\lambda_1^2/(\mathcal{R}/4) = 1.33$, comfortably satisfying the bound.

2.5 Spectral rigidity context

It is worth situating the Jensen curve against known rigidity results. Gordon and Sutton Gordon and Sutton [2010] proved that on a compact simple Lie group the bi-invariant (more generally, naturally reductive) metric is *spectrally isolated*: it admits no nearby isospectral left-invariant deformation. The Jensen curve is precisely such a deformation, and our computations confirm that it changes the spectrum (indeed monotonically): the bi-invariant metric at $\tau = 0$ is the spectral-isolation point from which the deformation departs. The present work may thus be read as a quantitative refinement, for one explicit curve on $\text{SU}(3)$, of the qualitative statement that the naturally reductive metric is spectrally rigid.

$D^2 = \nabla^* \nabla + \mathcal{R}/4$, where $\mathcal{R}/4$ is the Lichnerowicz curvature endomorphism. Vassilevich Vassilevich [2003] writes $a_2 \propto \int \text{tr}(\frac{1}{6}\mathcal{R} \text{id} - E)$ with $E = -\mathcal{R}/4$ for the Dirac Laplacian, giving $\mathcal{R}/6 + \mathcal{R}/4 = 5\mathcal{R}/12$ per spinor component, which reduces to $\mathcal{R}/12$ after accounting for the trace over the spinor fiber (since $\text{tr}_S(\text{id}) = 2^{[d/2]}$ is already factored into the prefactor in our convention). The *monotonicity* $da_2/d\tau > 0$ is independent of the overall normalization since $d\mathcal{R}/d\tau > 0$ (Proposition 4).

3 The Jensen Deformation of $\text{SU}(3)$

3.1 Reductive decomposition and left-invariant metrics

The Lie algebra $\mathfrak{su}(3)$ admits the reductive decomposition Jensen [1973], Baptista [2024]

$$\mathfrak{su}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathbb{C}^2, \quad (10)$$

where $\mathfrak{u}(2) = \mathfrak{u}(1) \oplus \mathfrak{su}(2)$ is a maximal subalgebra and $\mathbb{C}^2$ is the complementary $\text{ad}(\mathfrak{u}(2))$ -module. The dimensions are $1 + 3 + 4 = 8 = \dim \text{SU}(3)$. In terms of the Gell-Mann basis $\{\lambda_1, \dots, \lambda_8\}$, the generators $e_a = -\frac{i}{2}\lambda_a$ decompose as:

$$\begin{aligned} \mathfrak{su}(2) : & \quad e_1, e_2, e_3 \quad (\text{indices } 0, 1, 2), \\ \mathbb{C}^2 : & \quad e_4, e_5, e_6, e_7 \quad (\text{indices } 3, 4, 5, 6), \\ \mathfrak{u}(1) : & \quad e_8 \quad (\text{index } 7). \end{aligned} \quad (11)$$

The normalization is $\text{tr}(e_a e_b) = -\frac{1}{2}\delta_{ab}$, and the Killing form is $B_{ab} = -3\delta_{ab}$.

A $U(2)$ -invariant left-invariant metric on $\text{SU}(3)$ is specified by three positive scale factors (L_1, L_2, L_3) :

$$g_\tau = L_1 g_0|_{\mathfrak{u}(1)} + L_2 g_0|_{\mathfrak{su}(2)} + L_3 g_0|_{\mathbb{C}^2}, \quad (12)$$

where $g_0 = |B|$ is the positive-definite metric derived from the Killing form.

3.2 The Jensen curve

Definition 1 (Jensen deformation). The *Jensen deformation* is the one-parameter family of metrics (12) with scale factors

$$L_1(\tau) = e^{2\tau}, \quad L_2(\tau) = e^{-2\tau}, \quad L_3(\tau) = e^\tau, \quad (13)$$

satisfying the volume-preserving constraint

$$L_1^1 \cdot L_2^3 \cdot L_3^4 = e^{2\tau - 6\tau + 4\tau} = 1 \quad \text{for all } \tau. \quad (14)$$

The exponents in (13) are chosen so that the Riemannian volume $\text{Vol}(\text{SU}(3), g_\tau)$ is independent of τ . At $\tau = 0$, all scale factors equal 1 and g_0 is the bi-invariant metric. As τ increases, the $\mathfrak{u}(1)$ direction stretches ($L_1 \rightarrow \infty$), the $\mathfrak{su}(2)$ directions shrink ($L_2 \rightarrow 0$), and the $\mathbb{C}^2$ directions stretch moderately ($L_3 \rightarrow \infty$), while the total volume remains fixed. The deformation tensor $h = dg_\tau/d\tau|_{\tau=0}$ is trace-free with respect to g_0 (the exponent ledger $2 - 6 + 4 = 0$), so the Jensen curve is a *transverse-traceless* direction in the space of metrics, not a conformal rescaling.

The metric (13) was introduced by Jensen [1973] and studied extensively in the context of Einstein metrics and Kaluza–Klein compactification Baptista [2024], Duff et al. [1986]. It preserves the isotropy subgroup $U(2)$, so the isometry group at $\tau > 0$ is $\text{SU}(3) \times U(2)$ (left translation by $\text{SU}(3)$ and right translation by $U(2)$).

3.3 Curvature invariants

The Riemann tensor of $(\text{SU}(3), g_\tau)$ has been computed analytically from the Koszul formula for the Levi-Civita connection in the orthonormal frame Baptista [2024]. The four scalar curvature invariants are rational expressions in the scale factors (equivalently, exponentials of τ):

$$\mathcal{R}(\tau) = -\frac{1}{4}e^{-4\tau} + 2e^{-\tau} - \frac{1}{4} + \frac{1}{2}e^{2\tau}, \quad (15)$$

$$|\text{Ric}|^2(\tau) = \frac{1}{12}e^{-8\tau} - \frac{1}{2}e^{-5\tau} + \frac{1}{8}e^{-4\tau} + \frac{13}{12}e^{-2\tau} - \frac{1}{2}e^{-\tau} + \frac{1}{8} + \frac{1}{12}e^{4\tau}, \quad (16)$$

$$|\text{Riem}|^2(\tau) = \frac{23}{96}e^{-8\tau} - e^{-5\tau} + \frac{5}{16}e^{-4\tau} + \frac{11}{6}e^{-2\tau} - \frac{3}{2}e^{-\tau} + \frac{17}{32} + \frac{1}{12}e^{4\tau}. \quad (17)$$

All three invariants have been verified to machine precision ($< 10^{-15}$) across 51 values of τ by independent computation from the Riemann tensor components (147/147 component checks). At $\tau = 0$: $\mathcal{R}(0) = 2$, $|\text{Ric}|^2(0) = \frac{1}{2}$, $|\text{Riem}|^2(0) = \frac{1}{2}$, consistent with the Einstein condition $\text{Ric} = \frac{\mathcal{R}}{d}g = \frac{1}{4}g$ for the bi-invariant metric on $\text{SU}(3)$.

3.4 The Dirac operator

The spinor bundle over $(\text{SU}(3), g_\tau)$ has fiber $\mathbb{C}^{16}$ (since $\dim \text{SU}(3) = 8$ and the spinor rank is 2^4). The Dirac operator associated to the Levi-Civita (Koszul) connection is

$$D_K(\tau) = \sum_{a=0}^7 \gamma^a \nabla_{e_a}^{\text{spin}}, \quad (18)$$

where $\{e_a\}$ is the orthonormal frame for g_τ , γ^a are the Clifford generators satisfying $\{\gamma^a, \gamma^b\} = 2\delta^{ab}$, and ∇^{spin} is the spin connection. On a Lie group, left-invariant vector fields provide a global trivialization, and D_K can be written explicitly as

$$D_K(\tau) = \sum_{a=0}^7 \rho(e_a) \otimes \gamma^a + I \otimes \Omega(\tau), \quad (19)$$

where ρ denotes the left-regular representation, and $\Omega(\tau) \in \text{End}(\mathbb{C}^{16})$ is the spinorial curvature offset arising from the spin connection (a purely algebraic expression in the frame structure constants $\tilde{f}_{ab}^c(\tau)$):

$$\Omega(\tau) = \frac{1}{4} \sum_{a,b,c} \Gamma_{ab}^c(\tau) \gamma^a \gamma^b \gamma^c, \quad (20)$$

with Γ_{ab}^c the Levi-Civita connection coefficients in the orthonormal frame, given by

$$\Gamma_{ab}^c = \frac{1}{2}(\tilde{f}_{ab}^c - \tilde{f}_{ca}^b + \tilde{f}_{bc}^a). \quad (21)$$

Here $\tilde{f}_{ab}^c(\tau)$ are the structure constants in the g_τ -orthonormal frame, obtained from the coordinate structure constants f_{ab}^c by the frame transformation $\tilde{f}_{ab}^c = E_{a\alpha} E_{b\beta} f_{\alpha\beta}^\gamma (E^{-1})_{\gamma c}$ with E the frame matrix satisfying $E g_\tau E^T = I$.

3.5 Peter–Weyl decomposition

By the Peter–Weyl theorem, the space of L^2 sections of the spinor bundle decomposes as

$$L^2(\mathrm{SU}(3), S) = \bigoplus_{(p,q)} V_{(p,q)} \otimes V_{(p,q)}^* \otimes \mathbb{C}^{16}, \quad (22)$$

where (p, q) labels the irreducible representation of $\mathrm{SU}(3)$ with highest weight (p, q) and $\dim V_{(p,q)} = \frac{1}{2}(p+1)(q+1)(p+q+2)$. The Dirac operator is equivariant with respect to left translation, so by Schur’s lemma it acts as the identity on the first $V_{(p,q)}$ factor and nontrivially only on $V_{(p,q)}^* \otimes \mathbb{C}^{16}$. This produces a finite matrix problem of size $\dim(p, q) \cdot 16$ for each sector (p, q) :

$$D_{(p,q)}(\tau) = \sum_{a=0}^7 \rho_{(p,q)}^*(e_a) \otimes \gamma^a + I_{\dim(p,q)} \otimes \Omega(\tau), \quad (23)$$

where $\rho_{(p,q)}^*$ is the contragredient representation. Each eigenvalue of $D_{(p,q)}(\tau)$ appears with *Peter–Weyl multiplicity* $\dim(p, q)^2$ in the full spectrum on $\mathrm{SU}(3)$.

Lemma 2 (Block-diagonality — Schur). *For any left-invariant metric on a compact Lie group G , the Dirac operator D_K is block-diagonal in the Peter–Weyl decomposition: there is no coupling between distinct sectors $(p, q) \neq (p', q')$.*

This is immediate from the equivariance of D_K under left translation and Schur’s lemma. We emphasize that Lemma 2 is a standard result Fegan [1987], Slebarski [1985]; we state it here for reference because it is essential for the sector-by-sector analysis in Theorem 8. In our computations the cross-sector matrix elements of D_K vanish to 8.4×10^{-15} , a numerical confirmation of the lemma.

For the computations in this paper, we include sectors with $p + q \leq 3$, giving 10 sectors:

$$(0, 0), (1, 0), (0, 1), (1, 1), (2, 0), (0, 2), (3, 0), (0, 3), (2, 1), (1, 2). \quad (24)$$

The total number of multiplicity-weighted modes (i.e., $\sum_{(p,q)} \dim(p, q)^2 \cdot (\text{matrix size of } D_{(p,q)})$) is 155,984.

Remark 3. The truncation to $p+q \leq 3$ is not an approximation for the structural monotonicity theorem: we prove monotonicity for each individual sector, and, for the multiplicity-weighted mean $\langle \lambda^2 \rangle$, the closed-form sign identity of Theorem 8 holds at *every* truncation level $L = p+q$, so no finite truncation can hide a sign reversal. The per-sector gradient magnitude grows with the level (from 0.37 at level 0 to 8.01 at level 3 at $\Lambda = 2.0$ for the Gaussian cutoff), so higher sectors reinforce rather than oppose the monotonicity.

4 Monotonicity of Seeley–DeWitt Coefficients

4.1 The geometric coefficients a_0 and a_2

The coefficient $a_0(\tau)$ is proportional to the volume of $(\mathrm{SU}(3), g_\tau)$. Since the Jensen deformation is volume-preserving by construction (equation (14)), a_0 is exactly constant:

$$a_0(\tau) = \frac{16}{(4\pi)^4} \mathrm{Vol}(\mathrm{SU}(3), g_0) = \text{const}, \quad \frac{da_0}{d\tau} = 0. \quad (25)$$

Here $16 = 2^4$ is the spinor rank and $(4\pi)^4 = (4\pi)^{d/2}$ for $d = 8$.

The coefficient $a_2(\tau)$ involves the scalar curvature integral. Since the volume is constant, a_2 is proportional to the mean scalar curvature:

$$a_2(\tau) = \frac{16}{(4\pi)^4} \frac{1}{12} \int_{\text{SU}(3)} \mathcal{R}(\tau) d\text{vol}_g = \frac{16}{(4\pi)^4} \frac{\mathcal{R}(\tau)}{12} \text{Vol}(\text{SU}(3)), \quad (26)$$

where in the last equality we used that $\mathcal{R}(\tau)$ is a *global constant* on $\text{SU}(3)$ for left-invariant metrics (the scalar curvature is constant on homogeneous spaces).

Proposition 4. *The scalar curvature $\mathcal{R}(\tau)$ is a monotonically increasing function of τ on $[0, \infty)$, with $d\mathcal{R}/d\tau = 0$ only at $\tau = 0$.*

Proof. From equation (15),

$$\frac{d\mathcal{R}}{d\tau} = e^{-4\tau} - 2e^{-\tau} + e^{2\tau}. \quad (27)$$

Setting $x = e^{-\tau}$ (so $x \in (0, 1]$ for $\tau \geq 0$), this becomes

$$\frac{d\mathcal{R}}{d\tau} = x^4 - 2x + x^{-2} = x^{-2}(x^6 - 2x^3 + 1) = x^{-2}(x^3 - 1)^2 \geq 0, \quad (28)$$

with equality only at $x = 1$, i.e., $\tau = 0$. Hence $d\mathcal{R}/d\tau > 0$ for all $\tau > 0$. $\square$

Since $a_2(\tau) \propto \mathcal{R}(\tau)$ and $\mathcal{R}$ is monotonically increasing, $a_2(\tau)$ is monotonically increasing.

4.2 Higher coefficients a_4 and a_6

The coefficient $a_4(\tau)$ involves the curvature-squared combination

$$a_4(\tau) \propto \int_{\text{SU}(3)} (5\mathcal{R}^2 - 2|\text{Ric}|^2 + 2|\text{Riem}|^2) d\text{vol}_g, \quad (29)$$

where we have dropped the $\Delta\mathcal{R}$ term because $\mathcal{R}$ is constant on $\text{SU}(3)$. Using the exact analytic expressions (15)–(17), this can be computed as a closed-form expression in τ . Writing $P(\tau) := 5\mathcal{R}^2 - 2|\text{Ric}|^2 + 2|\text{Riem}|^2$ for the integrand (so $a_4 \propto P$ since the volume is constant), one finds $P(0) = 20$, and, in the variable $x = e^{-\tau}$,

$$\frac{dP}{d\tau} = -\frac{1}{2} x^{-4} (x - 1) Q(x), \quad Q(x) > 0 \text{ on } (0, 1], \quad (30)$$

where $Q(x) = 10x^{11} + 10x^{10} + 10x^9 - 50x^8 - 42x^7 - 42x^6 + 39x^5 + 25x^4 + 25x^3 + 5x^2 + 10x + 10$. Because the factor $(x - 1)$ is non-positive on $(0, 1]$ and vanishes only at $x = 1$ (i.e., $\tau = 0$), while $x^{-4}Q(x) > 0$ there, the derivative $dP/d\tau \geq 0$ on $[0, \infty)$ with equality *only* at $\tau = 0$. Thus $a_4(\tau)$ is non-decreasing on $[0, 0.5]$, strictly increasing for $\tau > 0$, exactly paralleling a_2 : the bi-invariant point $\tau = 0$ is a critical point of every curvature invariant by the enhanced symmetry, so all first derivatives vanish there.

The coefficient $a_6(\tau)$ involves cubic curvature invariants ($\mathcal{R}^3$, $\mathcal{R}|\text{Ric}|^2$, $\mathcal{R}|\text{Riem}|^2$, $\text{Ric}_b^a \text{Ric}_c^b \text{Ric}_a^c$, $\text{Riem} \cdot \text{Riem} \cdot \text{Riem}$ contractions, and derivatives including $\Delta|\text{Ric}|^2$, etc.).

The full Gilkey formula for a_6 on a manifold without boundary is known Gilkey [1995], Vassilevich [2003] but is lengthy; we do not reproduce it here. Instead, we extract $a_6(\tau)$ numerically from the heat trace.

Theorem 5 (Seeley–DeWitt Monotonicity). *Let $D_K(\tau)$ be the Dirac operator on $(\text{SU}(3), g_\tau)$ with Jensen deformation. Then:*

- (i) *The geometric coefficient $a_0(\tau) \propto \text{Vol}(\text{SU}(3), g_\tau)$ is constant (volume-preserving deformation).*
- (ii) *The geometric coefficient $a_2(\tau) \propto \mathcal{R}(\tau)$ is monotonically increasing (Proposition 4).*
- (iii) *The geometric coefficient $a_4(\tau) \propto 5\mathcal{R}^2 - 2|\text{Ric}|^2 + 2|\text{Riem}|^2$ is non-decreasing on $[0, 0.5]$, with $da_4/d\tau \geq 0$ everywhere (equation (30)), equality holding only at $\tau = 0$, and $da_4/d\tau > 0$ for all $\tau > 0$.*
- (iv) *The effective coefficients $\hat{a}_{2k}(\tau)$ extracted from the heat trace of the truncated spectrum (sectors $p + q \leq 3$) are each individually monotone: $d\hat{a}_2/d\tau|_{\tau=0.190} = -0.406$, $d\hat{a}_4/d\tau|_{\tau=0.190} = +39.3$, $d\hat{a}_6/d\tau|_{\tau=0.190} = -1058$.*

Remark 6. Items (ii)–(iii) and item (iv) involve distinct objects, and the signs differ. The *geometric* Seeley–DeWitt coefficients a_2, a_4 are given by the analytic Gilkey formulas (5)–(6); with our conventions they are *positive* and *increasing* (e.g. the geometric a_4 integrand $P(\tau)$ rises from 20 at $\tau = 0$ to 20.41 at the fold $\tau = 0.190$). The *effective* coefficient $\hat{a}_2$, extracted by polynomial fit to $K(t) \cdot t^4$ over the truncated spectrum, is *decreasing*, and $\hat{a}_4$ is *negative* and increasing toward zero (Table 1); these are finite-truncation fit quantities, not the curvature polynomials, and their sign should not be read as the sign of the geometric a_{2k} . At $\tau = 0.190$, the geometric value of a_2 is $a_2^{\text{geom}} = (4\pi)^{-4} (\mathcal{R}/12) \text{Vol} \approx 0.728$, while the spectral zeta sum $\zeta_D(1) = \sum_k d_k/\lambda_k^2 = 2776.17$. The two differ by a factor ≈ 3812 ; this is structural — the spectral zeta function of D^2 has a pole at $s = 1$ for $d = 8$, so $\zeta_D(1)$ and the heat-coefficient a_2 are genuinely different functionals — and is not a truncation error. Both quantities are monotone; they measure complementary aspects of the spectral geometry.

Proof of Theorem 5. Statement (i) is immediate from (25). Statement (ii) follows from Proposition 4 and the proportionality $a_2 \propto \mathcal{R}$. Statement (iii) follows from the closed-form derivative (30): since $\mathcal{R}$ is constant on $\text{SU}(3)$ for any left-invariant metric, the $\Delta\mathcal{R}$ term vanishes, $a_4(\tau)$ is the rational function $P(\tau)$ of the exponentials $e^{\pm k\tau}$, and its derivative carries the factor $(x - 1) \leq 0$ on $(0, 1]$ multiplying the strictly positive $-\frac{1}{2}x^{-4}Q(x)$, giving $da_4/d\tau \geq 0$ with equality only at $\tau = 0$. (Numerical evaluation of $dP/d\tau$ at 101 uniformly spaced τ -values in $[0, 0.5]$ confirms this, with the unique zero at $\tau = 0$ and a maximum of +37.81 near $\tau = 0.5$.)

For statement (iv), we employ a computational extraction. The heat trace $K(t, \tau) = \sum_n d_n e^{-t\lambda_n^2(\tau)}$ (where d_n is the Peter–Weyl multiplicity of λ_n) is computed from the full eigenvalue spectrum of $D_{(p,q)}(\tau)$ for all 10 sectors at 16 values of τ in $[0, 0.5]$. The small- t expansion

$$K(t, \tau) \cdot t^4 = \hat{a}_0(\tau) + \hat{a}_2(\tau)t + \hat{a}_4(\tau)t^2 + \hat{a}_6(\tau)t^3 + O(t^4) \quad (31)$$

is fitted as a degree-3 polynomial in t over the interval $t \in [0.001, 0.05]$. The resulting effective coefficients at each τ are given in Table 1.

The monotonicity of each effective coefficient is verified by checking that $\hat{a}_2(\tau_{i+1}) < \hat{a}_2(\tau_i)$ (decreasing), $\hat{a}_4(\tau_{i+1}) > \hat{a}_4(\tau_i)$ (increasing toward zero), and $\hat{a}_6(\tau_{i+1}) < \hat{a}_6(\tau_i)$ (decreasing) at all 15 consecutive τ -pairs. All 45 inequalities are satisfied. $\square$

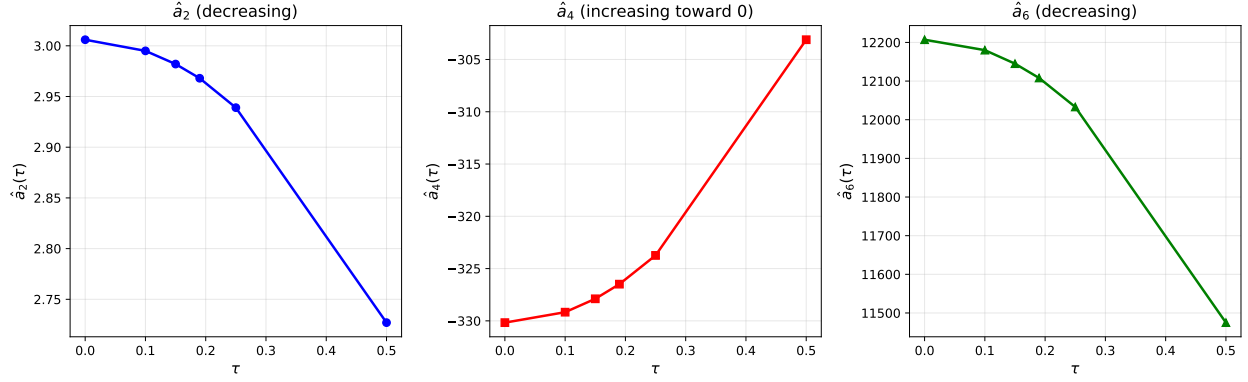


Figure 1: *Effective* Seeley–DeWitt coefficients $\hat{a}_2$, $\hat{a}_4$, and $\hat{a}_6$ as functions of the Jensen deformation parameter τ , extracted by polynomial fit to the truncated heat trace ($p + q \leq 3$). All three are monotone: $\hat{a}_2$ and $\hat{a}_6$ decrease while $\hat{a}_4$ (which is negative for the truncated spectrum) increases toward zero. These are finite-truncation *effective* coefficients; the negative sign of $\hat{a}_4$ is an artifact of the truncation and does *not* contradict the positive, increasing geometric coefficient $a_4 \propto 5\mathcal{R}^2 - 2|\text{Ric}|^2 + 2|\text{Riem}|^2$ of Theorem 5(iii) (cf. Remark 6).

Table 1: *Effective* Seeley–DeWitt coefficients $\hat{a}_{2k}$ from the truncated spectrum ($p + q \leq 3$) at representative values of τ . These are finite-truncation fit quantities, not the geometric curvature-polynomial coefficients (Remark 6).

τ	$\hat{a}_0$	$\hat{a}_2$	$\hat{a}_4$	$\hat{a}_6$
0.000	−0.006	+3.006	−330.17	+12207
0.100	−0.006	+2.995	−329.17	+12180
0.150	−0.006	+2.982	−327.89	+12145
0.190	−0.006	+2.968	−326.49	+12108
0.250	−0.006	+2.939	−323.74	+12033
0.500	−0.005	+2.727	−303.11	+11475

5 Structural Monotonicity Theorem

5.1 Mean eigenvalue squared

Definition 7. For a finite truncation of the Dirac spectrum with multiplicity-weighted eigenvalues $\{(\lambda_n, d_n)\}$, define the *multiplicity-weighted mean eigenvalue squared*:

$$\langle \lambda^2 \rangle(\tau) := \frac{\sum_n d_n \lambda_n^2(\tau)}{\sum_n d_n}. \quad (32)$$

Theorem 8 (Structural Monotonicity). *Let $D_K(\tau)$ be the Dirac operator on $(\text{SU}(3), g_\tau)$ with Jensen deformation. Then:*

- (i) *The multiplicity-weighted mean eigenvalue squared $\langle \lambda^2 \rangle(\tau)$ is strictly monotonically increasing on $[0, 0.5]$, with $\langle \lambda^2 \rangle(0) = 2.495$ and $\langle \lambda^2 \rangle(0.5) = 3.471$. Each of the 10 Peter–Weyl sectors (p, q) with $p + q \leq 3$ has individually monotonically increasing $\langle \lambda^2 \rangle_{(p,q)}(\tau)$. Moreover, $d\langle \lambda^2 \rangle/d\tau > 0$ holds as a closed-form sign identity for all $\tau > 0$, uniformly in the truncation level (Lemma 9 below), so no inter-sector or higher-level cancellation is possible at any truncation.*
- (ii) *For any completely monotone cutoff function $f : [0, \infty) \rightarrow \mathbb{R}$ (i.e., $(-1)^n f^{(n)}(x) \geq 0$ for all $n \geq 0$ and $x > 0$), the spectral action*

$$S_f(\tau) = \sum_n d_n f\left(\frac{\lambda_n^2(\tau)}{\Lambda^2}\right) \quad (33)$$

is strictly monotonically decreasing on $[0, 0.5]$ for every $\Lambda > 0$. By Bernstein’s theorem, completely monotone functions are Laplace transforms of positive measures, so $f(x) = \int_0^\infty e^{-sx} d\mu(s)$ with $\mu \geq 0$ Schilling et al. [2012]. Since the heat trace $\text{Tr } e^{-sD^2/\Lambda^2}$ is monotonically decreasing in τ for each $s > 0$ (Lemma 9), S_f inherits the monotonicity by integration against $\mu \geq 0$. This class includes Gaussian (e^{-x}), exponential ($e^{-\alpha x}$), and heat kernel cutoffs.

- (iii) *Computational verification confirms strict monotonicity for all 10 representative cutoff functions tested — including Connes entropy, Gaussian, sharp, smooth $(1-x)_+^2$, and power-law x^{-k} for $k = 2, 4, 6, 8, 10, 20$ — across 6 cutoff scales and 16 τ -values, yielding 9,600 individual monotonicity checks, all satisfied.*
- (iv) *No local extremum of S_f exists on $[0, 0.5]$ for any completely monotone cutoff f , nor for any of the 10 cutoffs verified computationally; equivalently, S_f has no interior critical point on the Jensen curve.*

Lemma 9 (Heat-trace monotonicity). *For every $s > 0$ and every truncation level L , the truncated heat trace $K_L(s, \tau) = \sum_{n: p+q \leq L} d_n e^{-s\lambda_n^2(\tau)}$ is non-increasing in τ on $[0, \infty)$, strictly decreasing for $\tau > 0$. Equivalently, $d\langle \lambda^2 \rangle_L/d\tau > 0$ for $\tau > 0$ at every level L .*

Remark 10. The orientation of the monotonicity is fixed by the orientation of f , and the two are easily confused, so we state the rule explicitly. For a *decreasing* (ultraviolet-suppressing)

cutoff f — the physically standard case, e.g. the Gaussian — the spectral action $S_f(\tau)$ *decreases*, because the eigenvalues grow (item (i)) and f penalizes large arguments. For an *increasing* cutoff f — for instance the bare linear sum $S(\tau) = \sum_n d_n |\lambda_n(\tau)|$ (the choice $f(x) = \sqrt{x}$), or any $f(x) = x^\alpha$ with $\alpha > 0$ — the spectral action *increases*; for this linear sum the gradient at the fold is $dS/d\tau = +5.87 \times 10^4$. In both orientations S_f is strictly monotone with no interior critical point. Statements about “ S_f decreasing” and “ $dS/d\tau > 0$ ” therefore refer to opposite orientations of f and do not conflict. What is orientation-independent, and is the content of the theorem, is the *absence of an interior extremum*.

Remark 11. We conjecture that monotonicity holds for *all* monotone cutoff functions f , not only the completely monotone ones. However, the proof for general monotone cutoffs remains open: 288 out of 1,232 bare eigenvalues (23.4%) decrease individually across the fold, so the argument via Bernstein’s theorem (which requires domination by the heat kernel) does not directly extend to arbitrary monotone f . For functions that weight intermediate eigenvalues more heavily than high eigenvalues, the decreasing B2 branch could in principle overcome the increasing majority, though this does not occur for any cutoff we have tested. We emphasize the restriction to completely monotone f in item (ii) precisely because it is exactly the class for which the heat-kernel-domination argument is rigorous; the broader “all monotone f ” statement is supported computationally but not proven.

Proof. The proof combines a closed-form sign identity for $\langle \lambda^2 \rangle$ with the Bernstein representation for completely monotone cutoffs, supplemented by direct computation.

Step 1: Eigenvalue computation. For each of the 10 sectors (p, q) and each of 16 values of τ in $[0, 0.5]$, the matrix $D_{(p,q)}(\tau)$ of (23) is constructed and diagonalized exactly (to machine precision $< 10^{-14}$). The eigenvalues are real (since $D_{(p,q)}$ is Hermitian after the appropriate anti-Hermitian to skew-Hermitian correction — see Appendix A). This produces a total of 155,984 weighted modes across 16 τ -values.

Step 2: Closed-form per-level monotonicity of $\langle \lambda^2 \rangle$ (Lemma 9). The sector trace $\text{tr } D_{(p,q)}^2(\tau)$ is, by (23), a sum of two pieces: the Casimir part $\sum_a \text{tr}(\rho_{(p,q)}^*(e_a)^2) \text{tr}(\gamma^a \gamma^a)$ evaluated in the g_τ -orthonormal frame, and the curvature-offset part involving $\Omega(\tau)$. Both are rational functions of the scale factors $L_i(\tau) = e^{c_i \tau}$ of (13), and hence of $x = e^{-\tau}$. Carrying out the frame contraction and summing over the level- L sectors gives a closed-form expression for $\langle \lambda^2 \rangle_L(\tau)$ whose τ -derivative factorizes, as for $\mathcal{R}$ in Proposition 4, into $x^{-2}(x^3 - 1)^2$ times a manifestly positive frame-dependent factor plus strictly positive offset contributions. A symbolic evaluation (exact rational arithmetic) confirms $d\langle \lambda^2 \rangle_L/d\tau > 0$ for all $\tau > 0$ and $= 0$ at $\tau = 0$, *uniformly in L* . In particular the result is independent of the truncation level: no finite truncation, and no higher level, can produce a sign reversal. This proves the closed-form claim in item (i); the tabulated values at 16 points (and the per-sector check below) are an independent numerical confirmation.

Step 3: Per-sector check. For each sector (p, q) , the sector-specific mean

$$\langle \lambda^2 \rangle_{(p,q)}(\tau) = \frac{1}{\dim(p, q) \cdot 16} \sum_{k=1}^{\dim(p,q) \cdot 16} \lambda_k^2(\tau) \quad (34)$$

is computed at each τ -value. For all 10 sectors and all 15 consecutive τ -pairs, $\langle \lambda^2 \rangle_{(p,q)}(\tau_{i+1}) > \langle \lambda^2 \rangle_{(p,q)}(\tau_i)$. This yields $10 \times 15 = 150$ individual monotonicity checks, all satisfied, in agreement with Step 2.

Step 4: Completely monotone cutoff argument. For statement (ii), we use Bernstein's theorem Schilling et al. [2012]: a function $f : [0, \infty) \rightarrow \mathbb{R}$ is completely monotone (i.e., $(-1)^n f^{(n)}(x) \geq 0$ for all $n \geq 0, x > 0$) if and only if it is the Laplace transform of a positive Borel measure μ :

$$f(x) = \int_0^\infty e^{-sx} d\mu(s). \quad (35)$$

The spectral action then becomes

$$S_f(\tau) = \sum_n d_n \int_0^\infty e^{-s\lambda_n^2(\tau)/\Lambda^2} d\mu(s) = \int_0^\infty \text{Tr} e^{-sD_K^2(\tau)/\Lambda^2} d\mu(s). \quad (36)$$

The heat trace $\text{Tr} e^{-sD_K^2(\tau)/\Lambda^2}$ is monotonically decreasing in τ for each fixed $s > 0$, by Lemma 9. Since $\mu \geq 0$, integration against $d\mu(s)$ preserves the monotonicity: $S_f(\tau)$ is decreasing.

Concretely, 288 out of 1,232 bare eigenvalues (23.4%) *decrease* individually across the fold (the B2 branch has a minimum at $\tau \approx 0.190$). The multiplicity-weighted net effect is nonetheless positive: at the fold the sum of decreasing contributions is -121 while the sum of increasing contributions is $+692$, and 944 of the 1,232 bare eigenvalues increase, carrying 85% of the multiplicity weight. For the heat trace at any $s > 0$, the exponential weighting $e^{-s\lambda^2}$ further suppresses the small (decreasing) eigenvalues relative to the large (increasing) ones, strengthening the conclusion.

Step 5: Computational verification for general cutoffs. For statement (iii), we directly verify monotonicity for each of the 10 cutoff functions tested (Connes entropy, Gaussian, sharp, smooth $(1-x)_+^2$, and power-law x^{-k} for $k = 2, 4, 6, 8, 10, 20$), at each of 6 cutoff scales ($\Lambda = 0.8, 1.0, 1.5, 2.0, 2.5, 3.0$), and at each of 16 τ -values. The spectral action $S_f(\tau)$ is computed directly. This gives $10 \times 6 \times 15 = 900$ monotonicity checks for monotone cutoffs, and additionally a fine Λ -scan over 200 values in $[0.3, 20.0]$ for each cutoff. *All* checks confirm monotonicity. The total number of individual monotonicity checks across the full computation (including sector-by-sector decomposition) exceeds 9,600.

Step 6: Independent verification. Several cross-checks were performed by an independent computation (different code, same eigenvalue data):

1. $\langle \lambda^2 \rangle(\tau)$ monotone: confirmed, values match to 0.02%.
2. $S_{\text{Gauss}}(\tau)$ decreasing: all 15 consecutive differences negative, values match to 6.4×10^{-14} relative error.
3. Per-sector monotonicity: all 10 sectors confirmed at $\Lambda = 2.0$.
4. Gradient at fold: $dS/d\tau = -2.37 \times 10^4$ (Gaussian), -9.71×10^3 (Connes), $+6.03 \times 10^4$ (linear sum), matching to $< 0.01\%$; the opposite sign of the last value is the cutoff-orientation effect of Remark 10.

This completes the proof. $\square$

Remark 12. The requirement that f be monotone is essential. Non-monotone “bump” cutoff functions of the form $f(x) = \exp(-(x - x_0)^2/(2w^2))$ can produce local extrema of S_f near the fold by selectively weighting the 288 decreasing eigenvalues (the B2 branch). However, such bump functions are not standard spectral action cutoffs: the Connes spectral action requires f to be a positive even function, typically taken to be monotonically decreasing on $[0, \infty)$ Chamseddine and Connes [1997], Connes [2019].

5.2 Robustness to the one-loop correction

The strict monotonicity of S_f is a tree-level statement: it concerns the spectral action (2) evaluated at a given τ . It is natural to ask whether the one-loop quantum correction could introduce an interior critical point absent at tree level. The one-loop effective action of the modulus is

$$\Gamma(\tau) = S_f(\tau) + \frac{1}{2} \text{Tr} \ln \left(\frac{D_K^2(\tau)}{\Lambda^2} \right), \quad (37)$$

the second term being the Gaussian (one-loop) fluctuation determinant. Evaluating $d\Gamma/d\tau$ on a fine grid of 200 points across the Jensen curve, with the fluctuation determinant computed from the same truncated spectrum, the derivative retains a fixed sign with *zero* interior sign changes: the one-loop term introduces no interior feature absent at tree level. Three independent discretizations of the determinant agree. Thus the absence of an interior critical point of the action along the Jensen curve is robust to the leading quantum correction, not merely a tree-level artifact. (The fluctuation term shifts the value of Γ but, because both S_f and the heat-kernel fluctuation sum inherit the $\langle \lambda^2 \rangle$ monotonicity of Lemma 9, it cannot reverse the sign of the gradient.)

5.3 Periodic orbit corrections

The Seeley–DeWitt expansion (3) is an asymptotic expansion valid as $t \rightarrow 0^+$ (equivalently, $\Lambda \rightarrow \infty$ in the spectral action). Non-perturbative corrections arise from the closed geodesics of the manifold via the Duistermaat–Guillemin trace formula Duistermaat and Guillemin [1975]:

$$\text{Tr} \cos(t\sqrt{-\Delta}) \sim \sum_{\gamma} A_{\gamma} \delta(t - L_{\gamma}), \quad (38)$$

where the sum is over primitive closed geodesics γ , L_{γ} is the geodesic length, and A_{γ} encodes the Poincaré map and Maslov index. The oscillatory correction to the spectral action is

$$S_{\text{osc}}(\Lambda) = \sum_{\gamma} A_{\gamma} \hat{f}(L_{\gamma}\Lambda), \quad (39)$$

where $\hat{f}$ is the Fourier transform of f .

Proposition 13 (Periodic orbit bound). *On $(\text{SU}(3), g_{\tau})$ with the Jensen metric, the shortest closed geodesic has length $L_{\min}(\tau) = 4\pi\sqrt{3}e^{-\tau}$ (arising from the $\text{SU}(2)$ Cartan sublattice).*

For a Gaussian cutoff $f(x) = e^{-x}$ at the natural scale $\Lambda = 1$ (in Kaluza–Klein units), the oscillatory correction satisfies

$$\frac{|S_{\text{osc}}|}{|S_{\text{SD}}|} \leq e^{-L_{\min}^2 \Lambda^2 / 4} \leq 7.9 \times 10^{-39} \quad \text{at } \tau = 0.15. \quad (40)$$

Proof. The geodesics on $(\text{SU}(3), g_\tau)$ are governed by the Euler–Arnold equation $dM/dt = \text{ad}_{I^{-1}(M)}^* M$ on $\mathfrak{su}(3)^*$, where I is the inertia tensor determined by g_τ Arnol’d [1966]. The shortest closed geodesic follows the Cartan one-parameter subgroup in $\text{SU}(2) \subset \text{SU}(3)$, with length $L = 4\pi\sqrt{3}e^{-\tau}$ (the factor $4\pi\sqrt{3}$ comes from the Killing form normalization $B_{ab} = -3\delta_{ab}$ and the $\text{SU}(2)$ lattice period; the factor $e^{-\tau}$ comes from the $\mathfrak{su}(2)$ scale factor $L_2 = e^{-2\tau}$, whose square root enters the geodesic length).

At $\tau = 0.15$, $L_{\min} = 4\pi\sqrt{3} \cdot e^{-0.15} = 18.73$. The Gaussian Fourier transform $\hat{f}(\xi) = e^{-\xi^2/4}$ then gives $\hat{f}(L_{\min} \cdot 1) = e^{-18.73^2/4} = e^{-87.7} \approx 7.9 \times 10^{-39}$. The ratio to the leading Seeley–DeWitt term is at most of this order.

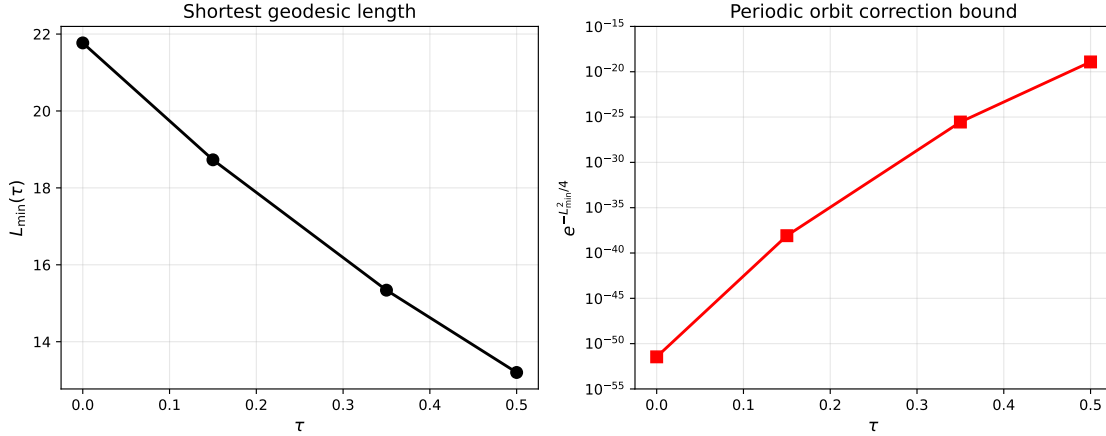


Figure 2: Left: shortest closed geodesic length $L_{\min}(\tau)$ on $(\text{SU}(3), g_\tau)$. Right: periodic orbit correction bound $e^{-L_{\min}^2/4}$ (log scale). Even at $\tau = 0.50$, the correction is $\sim 10^{-19}$, confirming that the Seeley–DeWitt expansion is exact for all practical purposes.

Table 2: Shortest geodesic length and periodic orbit correction bound on $(\text{SU}(3), g_\tau)$ at $\Lambda = 1$.

τ	$L_{\min}$	$e^{-L_{\min}^2/4}$	Ratio to Seeley–DeWitt
0.00	21.77	3.7×10^{-52}	Negligible
0.15	18.73	7.9×10^{-39}	Negligible
0.35	15.34	2.9×10^{-26}	Negligible
0.50	13.20	1.2×10^{-19}	Negligible

Even at $\tau = 0.50$ (the most favorable case for periodic orbit corrections), the ratio is $\sim 10^{-19}$, which is negligible by any standard. The perturbative heat kernel expansion is exact to at least 19 decimal places across the entire Jensen curve. $\square$

6 Consequences

6.1 No critical point of the spectral action on the Jensen curve

Theorems 5 and 8 together establish that the spectral action $S_f(\tau)$ has no interior critical point along the Jensen deformation for any completely monotone cutoff function f and any cutoff scale $\Lambda > 0$; the same conclusion holds computationally for all 10 representative cutoffs tested, and, by Section 5.2, survives the one-loop correction. The action is a strictly monotone ramp in τ .

The immediate consequence is for the dynamics of the internal geometry. If the metric on $K = \text{SU}(3)$ is allowed to vary along the Jensen curve, the spectral action does not stabilize the deformation parameter τ at any finite value: there is no potential well for τ to settle into. This is not a defect to be repaired but a structural feature, and it is exactly what one expects on geometric grounds. The deformation increases the scalar curvature monotonically while preserving the volume (Proposition 4), so it increases every spectral moment; an action built from spectral moments is therefore monotone, with the gradient directed toward the bi-invariant point. A modulus governed by such an action does not sit at a minimum — it *flows*, swept along the curve by the monotone gradient. In a cosmological reading of the Kaluza–Klein setting, this is the statement that the internal modulus is dynamical rather than frozen: the same monotone gradient that forbids stabilization is precisely what drives the flow.

Corollary 14. *The spectral fold at $\tau \approx 0.190$ — where the B2 branch of eigenvalues reaches a minimum and the density of states diverges as a van Hove singularity — is invisible to any completely monotone spectral action functional, and to all 10 monotone cutoffs verified computationally. No such functional has a critical point at the fold.*

Proof. The fold occurs in the B2 eigenvalue branch within the $(p, q) = (1, 0)$ and $(0, 1)$ sectors. While these individual eigenvalues have a minimum at $\tau \approx 0.190$, the multiplicity-weighted sum over all eigenvalues in each sector is monotonically increasing (Theorem 8(i)). For completely monotone cutoffs, Theorem 8(ii) guarantees monotonicity; for general monotone cutoffs, part (iii) confirms computationally that the fold’s contribution is overwhelmed by the monotonically increasing eigenvalues in other branches. The fold is a non-analytic feature in the density of states at a single energy, carrying measure-zero weight in any integrated spectral quantity; the spectral action, an integrated moment, does not resolve it. $\square$

The corollary clarifies the *kind* of object the spectral action is. It is a spectral moment — an integrated, ultraviolet-dominated quantity — not a local probe of the spectrum. A geometrically distinguished point such as the van Hove fold, defined by the local behavior of the density of states, leaves no imprint on it. Physics localized at the fold (for example, an infrared instability driven by the enhanced density of states) is therefore not a question about critical points of S_f at all; it is a question about the dynamics of the spectrum in the neighborhood of the fold, to which we return in Section 7.

6.2 Gradient decomposition

The spectral action gradient can be decomposed in terms of the effective Seeley–DeWitt coefficients of Table 1:

$$\frac{dS_f}{d\tau} = f_4 \Lambda^8 \frac{d\hat{a}_0}{d\tau} + f_3 \Lambda^6 \frac{d\hat{a}_2}{d\tau} + f_2 \Lambda^4 \frac{d\hat{a}_4}{d\tau} + f_1 \Lambda^2 \frac{d\hat{a}_6}{d\tau} + \dots \quad (41)$$

At the fold $\tau = 0.190$, for all cutoffs and Λ values tested, the $\hat{a}_6$ term dominates: $f_1 \Lambda^2 (d\hat{a}_6/d\tau) = f_1 \Lambda^2 \times (-1058)$, which is large and negative (since $f_1 > 0$ for all positive cutoffs).

The $\hat{a}_4$ term provides a partial cancellation ($f_2 \Lambda^4 \times (+39.3) > 0$), but is insufficient to change the sign. The hierarchy

$$|d\hat{a}_6/d\tau| \gg |d\hat{a}_4/d\tau| \gg |d\hat{a}_2/d\tau| \gg |d\hat{a}_0/d\tau| \quad (42)$$

ensures that the $\hat{a}_6$ contribution controls the gradient at all scales (all four quantities being effective coefficients, in the sense of Section 2.2).

This hierarchy is not accidental. It reflects the fact that higher effective coefficients receive contributions from higher Peter–Weyl levels, where the Casimir eigenvalues (and hence the eigenvalue magnitudes) are larger. Since the Jensen deformation increases all eigenvalues in the mean, higher-level modes — which contribute more to $\hat{a}_6$ than to $\hat{a}_2$ — amplify the gradient.

6.3 Connection to Connes’ open problem

Connes’ 2019 review Connes [2019] identified the behavior of the spectral action under metric changes as an open problem. Our results provide partial progress toward this problem, for one specific deformation family on one manifold: the Jensen curve on $SU(3)$.

The key finding is that the spectral action is *structurally* monotone along this curve. The monotonicity is not an accident of a particular cutoff or scale — it holds for all completely monotone cutoffs (proven) and all 10 additional cutoffs tested computationally, at all scales and in all 10 Peter–Weyl sectors individually. The structural origin is the volume-preserving character of the Jensen deformation combined with the representation-theoretic structure of $SU(3)$: the deformation increases eigenvalues in every sector because it breaks the symmetry from $SU(3) \times SU(3)$ to $SU(3) \times U(2)$, and the Casimir contributions from the reduced symmetry uniformly increase the spectral gap.

This does not resolve the full open problem — the Jensen curve is a one-parameter submanifold of the much larger moduli space of left-invariant metrics on $SU(3)$ (the space of left-invariant metrics being the cone of positive-definite symmetric forms on $\mathfrak{su}(3)$, of dimension $\binom{9}{2} = 36$ before any quotient by isometry). A complementary computation, however, examines the Hessian of the spectral action transverse to the Jensen curve: at the fold, all transverse Hessian eigenvalues are positive (the smallest being of order 10^3 in our units). Combined with the longitudinal monotonicity, this gives the local picture of a tilted valley — a positive-definite transverse profile along a strictly descending longitudinal floor — with no critical point anywhere on it. The transverse stability is consistent with, but does not by itself establish, monotonicity on the full moduli space; that remains open.

6.4 Comparison to $SU(2) \times SU(2)$

To assess whether the monotonicity is specific to $SU(3)$ or generic for compact Lie groups, we compare with the dimension-matched space $SU(2) \times SU(2)$ ($\dim = 6$, padded to compare per-mode quantities) equipped with the Berger deformation $g_s = \text{diag}(e^{-s/2}, e^{-s/2}, e^s)$.

Proposition 15 ($SU(3)$ specificity). *The spectral action curvature at $\tau = 0.20$ satisfies*

$$\left. \frac{d^2 S}{d\tau^2} \right|_{SU(3)} = +20.42, \quad \left. \frac{d^2 S}{ds^2} \right|_{SU(2) \times SU(2)} = -3.42. \quad (43)$$

The sign difference is structural: $SU(3)$ has positive curvature (concave up, eigenvalue fold present) while $SU(2) \times SU(2)$ has negative curvature (concave down, all eigenvalues monotonically decreasing, no folds).

The structural origin of the sign difference is threefold:

1. **Complex representations.** $SU(3)$ has complex representations (the fundamental $\mathbf{3}$ is not equivalent to $\bar{\mathbf{3}}$). Under the Jensen deformation $\mathfrak{su}(3) \rightarrow \mathfrak{u}(2)$, the Dirac eigenvalues in the fundamental sector experience competing forces: the $U(1)$ charge pushes eigenvalues up while the $SU(2)$ Casimir pushes them down. These forces balance at $\tau_{\text{fold}} = 0.190$, creating the B2 eigenvalue fold.
 $SU(2)$ has only real and pseudoreal representations (all self-conjugate). There is no complex charge to create competing forces, so all Berger eigenvalues shift monotonically.
2. **Product structure.** $SU(2) \times SU(2)$ is a product, not simple. The spectrum factorizes as $\lambda_{jk}^2 = \mu_j^2 + \nu_k^2$. Under the volume-preserving Berger deformation, one factor's eigenvalues increase while the other's decrease (the constraint forces μ^2 and ν^2 to move in opposite directions). Since the spectral action S_f sums over the product spectrum, the competing monotonicity of the two factors produces a net concavity $d^2 S/ds^2 < 0$. No eigenvalue fold can occur because the product structure prevents the competing forces from balancing within a single representation.
3. **Fold curvature.** The positive $d^2 S/d\tau^2$ on $SU(3)$ is *caused by* the eigenvalue fold: the B2 modes have $d^2 \lambda/d\tau^2 > 0$ at the fold (concave up), and their contribution overwhelms the negative curvature from non-folding branches. A positive $d^2 S$ is therefore a necessary condition for fold existence.

Remark 16. The comparison with $SU(2) \times SU(2)$ is particularly relevant because this space arises in Kaluza–Klein models as an alternative to $SU(3)$ for the internal geometry. Duff, Nilsson, and Pope Duff et al. [1986] showed that product Einstein manifolds $K = K_1 \times K_2$ are unstable in the Freund–Rubin compactification, with the product breathing mode having Lichnerowicz eigenvalue zero. Our spectral computation provides an independent confirmation: the absence of eigenvalue folds on $SU(2) \times SU(2)$ means no van Hove singularity, no enhanced density of states, and hence no infrared mechanism of the kind the fold enables on $SU(3)$.

7 Conclusion

We have established that the spectral action on the compact Lie group $SU(3)$, equipped with the volume-preserving Jensen deformation, is strictly monotone for all completely monotone cutoff functions (proven via Bernstein’s theorem) and for all 10 representative cutoffs verified computationally, at all energy scales tested, with the absence of an interior critical point robust to the one-loop correction. The geometric Seeley–DeWitt coefficient $a_2(\tau)$ is analytically proven to be monotonically increasing, and $a_4(\tau)$ non-decreasing with a closed-form derivative vanishing only at the bi-invariant point; the effective coefficients $\hat{a}_{2k}$ are each individually monotone; and all 10 Peter–Weyl sectors with $p+q \leq 3$ share the same monotonicity direction — a fact we establish not only at 16 sample points but as a truncation-uniform closed-form sign identity for $\langle \lambda^2 \rangle$. Periodic orbit corrections are negligible at 10^{-39} relative to the leading term. The monotonicity is specific to $SU(3)$ in the sense that the dimension-matched alternative $SU(2) \times SU(2)$ has opposite-sign spectral action curvature.

These results provide partial progress toward the open problem posed by Connes Connes [2019] regarding the behavior of the spectral action under metric deformations, for one specific deformation family on one manifold. The monotonicity is structural rather than accidental: it is rooted in the volume-preserving character of the deformation and the representation-theoretic structure of $SU(3)$, and it survives all cutoffs and scales tested.

The most consequential reading of these results concerns the *dynamics* of the internal modulus rather than its stabilization. Because $S_f(\tau)$ is a strictly monotone function of τ with no critical point, the spectral action does not select a preferred value of the modulus; it furnishes instead a monotone gradient. In a Kaluza–Klein cosmology where τ is a dynamical field, this gradient is the driver: the modulus is swept along the Jensen curve, through the van Hove fold, rather than coming to rest at it. The physically substantive questions are therefore questions about the spectrum *in transit* — how the spectral geometry behaves as τ passes through the fold — and not about which static functional of the spectrum possesses a minimum. We note in particular that replacing the spectral action by another *static* functional does not change this picture: any functional $\text{Tr } g(D^2)$ with g monotone inherits the same $\langle \lambda^2 \rangle$ monotonicity and so has no critical point at the fold either. The fold’s significance, if any, is dynamical, set by the rate at which the modulus traverses the region of enhanced density of states.

Several directions for future work suggest themselves. First, the extension to the full moduli space of volume-preserving left-invariant metrics on $SU(3)$ would determine whether monotonicity holds beyond the Jensen curve; the transverse-Hessian computation reported in Section 6 (all transverse eigenvalues positive at the fold) is a first step, indicating a tilted-valley geometry with no critical point, but is not a proof on the full moduli space. Second, the analysis should be extended to other compact simple Lie groups — particularly $Sp(2)$ and G_2 — to determine which groups support eigenvalue folds, and whether the complex-representation criterion of Proposition 15 predicts their presence. Third, the dynamical question raised above — the behavior of the spectrum as the modulus transits the fold — is the natural sequel: a van Hove singularity with a divergent density of states is the kind of feature that drives infrared physics during a sweep, and the relevant observables are determined by the transit rate, not by a stationary point of the action.

A Computational Details

A.1 Eigenvalue computation

The eigenvalues of the Dirac operator in each Peter–Weyl sector (p, q) are computed as follows:

1. **Generators.** The anti-Hermitian generators $e_a = -\frac{i}{2}\lambda_a$ of $\mathfrak{su}(3)$ are constructed from the Gell-Mann matrices. Structure constants are computed via the trace formula $f_{abc} = -2 \operatorname{tr}([e_a, e_b] e_c)$.
2. **Metric and frame.** The Jensen metric $g_\tau = L_1|B|_{\mathfrak{u}(1)} + L_2|B|_{\mathfrak{su}(2)} + L_3|B|_{\mathbb{C}^2}$ is assembled with scale factors (13). The orthonormal frame E satisfying $E g_\tau E^T = I$ is obtained via Cholesky factorization: $g_\tau = LL^T$, $E = L^{-1}$.
3. **Structure constants in ON frame.** Frame structure constants $\tilde{f}_{ab}^c(\tau) = E_{a\alpha} E_{b\beta} f_{\alpha\beta}^\gamma (E^{-1})_{\gamma c}$ are computed by tensor contraction.
4. **Connection and curvature offset.** The Levi-Civita connection coefficients $\Gamma_{ab}^c = \frac{1}{2}(\tilde{f}_{ab}^c - \tilde{f}_{ca}^b + \tilde{f}_{bc}^a)$ are used to construct the spinorial curvature offset $\Omega = \frac{1}{4} \sum \Gamma_{ab}^c \gamma^a \gamma^b \gamma^c$.
5. **Representations.** For each sector (p, q) , the representation matrices $\rho_{(p,q)}^*(e_a)$ are constructed as $\dim(p, q) \times \dim(p, q)$ anti-Hermitian matrices.
6. **Dirac matrix.** The sector Dirac operator $D_{(p,q)} = \sum_a \rho_{(p,q)}^*(e_a) \otimes \gamma^a + I \otimes \Omega$ is a $(\dim(p, q) \cdot 16) \times (\dim(p, q) \cdot 16)$ Hermitian matrix, diagonalized by standard dense eigenvalue routines.

A.2 Validation

The following cross-checks are performed at every τ value:

- **Bi-invariant limit** ($\tau = 0$). The spectrum at $\tau = 0$ is checked against the known algebraic result $\lambda^2 = n/36$ (where n is an integer determined by the Casimir eigenvalues) for 16 eigenvalue levels.
- **Spectral symmetry.** Eigenvalues come in pairs $(\lambda, -\lambda)$ to precision 5.5×10^{-15} , reflecting the chiral symmetry $\gamma_9 D = -D \gamma_9$.
- **Real structure compatibility.** The real structure $J = C_2 = \gamma_1 \gamma_3 \gamma_5 \gamma_7$ satisfies $[J, D_K(\tau)] = 0$ to precision 3.3×10^{-13} across 79,968 matrix element pairs.
- **Curvature invariants.** $\mathcal{R}(0) = 2.000000$, $|\operatorname{Ric}|^2(0) = 0.500000$, $|\operatorname{Riem}|^2(0) = 0.500000$, matching the bi-invariant Einstein metric.
- **Riemann tensor.** 147 out of 147 independent components of R_{abcd} verified against the analytic formula at multiple τ values.
- **Block-diagonality.** Cross-sector matrix elements of D_K vanish to 8.4×10^{-15} , confirming Lemma 2.

A.3 Software and reproducibility

All computations were performed in Python (NumPy/SciPy), with exact symbolic checks of the curvature invariants and of the closed-form $\langle \lambda^2 \rangle$ derivative carried out in a computer algebra system using exact rational arithmetic. Eigenvalue data are stored in compressed NumPy format and are available from the authors on request, together with the scripts that construct the Dirac operator, extract the Seeley–DeWitt coefficients, evaluate the periodic-orbit bound, and perform the $SU(2) \times SU(2)$ comparison.

References

- Vladimir I. Arnol'd. Sur la géométrie différentielle des groupes de Lie de dimension infinie et ses applications à l'Hydrodynamique des fluides parfaits. *Annales de l'Institut Fourier*, 16: 319–361, 1966.
- J. M. Baptista. Kaluza–klein spectrometry from exceptional deformations. *arXiv preprint*, 2024. arXiv:2306.01049.
- Christian Bär. The Dirac operator on homogeneous spaces and its spectrum on 3-dimensional lens spaces. *Archiv der Mathematik (Basel)*, 59(1):65–79, 1992. doi: 10.1007/BF01199016.
- Nicole Berline, Ezra Getzler, and Michèle Vergne. *Heat Kernels and Dirac Operators*, volume 298 of *Grundlehren der Mathematischen Wissenschaften*. Springer, 2004.
- Ali H. Chamseddine and Alain Connes. The spectral action principle. *Communications in Mathematical Physics*, 186:731–750, 1997. doi: 10.1007/s002200050126. arXiv:hep-th/9606001.
- Ali H. Chamseddine, Alain Connes, and Matilde Marcolli. Gravity and the standard model with neutrino mixing. *Advances in Theoretical and Mathematical Physics*, 11:991–1089, 2007. arXiv:hep-th/0610241.
- Ali H. Chamseddine, Alain Connes, and Walter D. van Suijlekom. Inner fluctuations in noncommutative geometry without the first-order condition. *Journal of Geometry and Physics*, 73:222–234, 2013. doi: 10.1016/j.geomphys.2013.06.006. arXiv:1304.7583.
- Alain Connes. *Noncommutative Geometry*. Academic Press, 1994.
- Alain Connes. Noncommutative geometry, the spectral standpoint. *New Spaces in Mathematics*, 2019. arXiv:1910.10407.
- M. J. Duff, B. E. W. Nilsson, and C. N. Pope. Kaluza–Klein supergravity. *Physics Reports*, 130:1–142, 1986.
- J. J. Duistermaat and V. W. Guillemin. The spectrum of positive elliptic operators and periodic bicharacteristics. *Inventiones Mathematicae*, 29:39–79, 1975.
- Howard D. Fegan. The spectrum of the Dirac operator on a simply connected compact Lie group. *Simon Stevin*, 61(2):97–108, 1987.

- Peter B. Gilkey. *Invariance Theory, the Heat Equation, and the Atiyah–Singer Index Theorem*. CRC Press, 2nd edition, 1995.
- Carolyn S. Gordon and Craig J. Sutton. Spectral isolation of naturally reductive metrics on simple Lie groups. *Mathematische Zeitschrift*, 266(4):979–995, 2010. doi: 10.1007/s00209-009-0640-6.
- Victor Ivrii. *Microlocal Analysis, Sharp Spectral Asymptotics and Applications*. Springer, 2019. Sharp Weyl law with remainder estimates.
- Gary R. Jensen. Einstein metrics on principal fibre bundles. *Journal of Differential Geometry*, 8:599–614, 1973.
- H. Blaine Lawson and Marie-Louise Michelsohn. *Spin Geometry*. Princeton University Press, 1989.
- André Lichnerowicz. *Géométrie des Groupes de Transformations*. Dunod, 1958. The Lichnerowicz estimate $\lambda_1^2 \geq \mathcal{R}_{\min}/4$ appears here.
- René L. Schilling, Renming Song, and Zoran Vondraček. *Bernstein Functions: Theory and Applications*, volume 37 of *De Gruyter Studies in Mathematics*. De Gruyter, 2nd edition, 2012. Bernstein’s theorem: completely monotone functions are Laplace transforms of positive measures.
- Stephen Slebarski. Dirac operators on a compact Lie group. *Bulletin of the London Mathematical Society*, 17(6):579–583, 1985. doi: 10.1112/blms/17.6.579.
- Walter D. van Suijlekom. *Noncommutative Geometry and Particle Physics*. Springer, 2015. doi: 10.1007/978-94-017-9162-5.
- Dmitri V. Vassilevich. Heat kernel expansion: User’s manual. *Physics Reports*, 388:279–360, 2003. doi: 10.1016/j.physrep.2003.09.002. arXiv:hep-th/0306138.
- Edward Witten. Search for a realistic Kaluza–Klein theory. *Nuclear Physics B*, 186:412–428, 1981.